

PretrainLab-X: Day 5 Final Polish

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A production-shaped LLaMA-style foundation-model pretraining stack with real FineWeb-Edu data, BF16 training, activation-checkpointing comparison, throughput measurement, adaptive health monitoring, and reproducible experiment artifacts.

Scope boundary

These are controlled validation runs. The Day 5 suite includes a 336M-parameter run targeted at approximately 500M tokens, plus bounded memory and throughput experiments. This remains a small-scale pretraining validation, not a claim of completing the 3B/5B-token Day 4 targets.

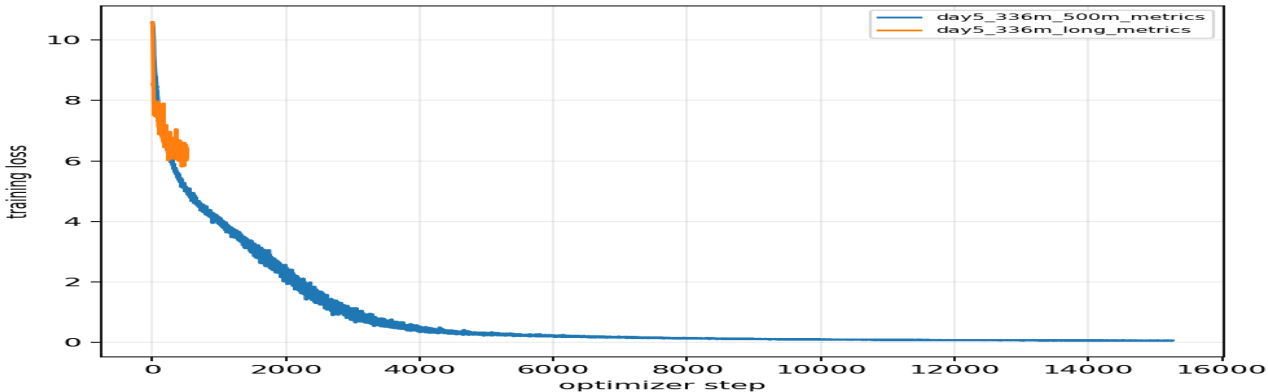
1. Architecture and training system

The model exposes RMSNorm, RoPE, SwiGLU, grouped-query attention, PyTorch SDPA, tied/untied embeddings, and activation checkpointing. The engine provides BF16 autocast, gradient accumulation, clipping, warmup-cosine scheduling, atomic checkpoint/resume, JSONL metrics, optional W&B; logging, and GPU snapshots.

Component	Evidence
Data	FineWeb-Edu sample-10BT slice; 10M tokens; deterministic filters and exact-text dedup
Precision	BF16 autocast on CUDA
Memory	Activation checkpointing switch and paired on/off runs
Observability	Loss, LR, gradient norm, throughput, GPU memory/utilization, health status
Recovery	Atomic checkpoint with optimizer, scheduler, RNG and data provenance

2. Recorded training runs

Run	Parameters	Steps	Tokens	Final loss	Val loss	Runtime (s)
336M 500M-token	336,118,784	15259	500006912	0.0693	-	12673.9
336M bounded	336,118,784	512	1048576	6.2389	5.9079	237.5
653M checkpoint off	653,477,120	3	1536	9.8340	9.3271	9.0
653M checkpoint on	653,477,120	3	1536	9.8344	9.3284	7.5



Observed loss curves; short runs are not capability evaluations.

3. Efficiency and memory trade-off

Throughput benchmark

Batch	Seq	Status	Tokens/s	Peak memory (MB)
1	512	ok	5333.3	6782.1
2	512	ok	14171.5	6836.7
4	512	ok	29144.4	8276.8
8	512	ok	47667.1	11803.2

Activation-checkpointing comparison

Mode	Steps	Peak memory (MB)	Mean tokens/s	Status
checkpoint off	3	12836.1	608.9	ok
checkpoint on	3	12795.4	552.7	ok

The intended engineering trade-off is lower activation memory in exchange for extra recomputation; the table reports only what the recorded run measured.

4. Adaptive training health

```
{ "steps": 512, "status": "warning", "warning_steps": [ 5, 7, 180, 232, 245, 423, 433, 443, 453, 463, 473 ], "reasons": [ "gradient_spike", "low_gpu_utilization" ], "max_grad_norm": 10.946956634521484, "mean_tokens_per_sec": 4389.0570174007335, "mean_gpu_utilization_pct": 32.552734375 }
```

5. W&B; reproducibility

Training logs are mirrored to W&B; offline run directories when the SDK is available. The same metrics remain in JSONL so the experiment is auditable without a network connection.

Run	Mode	Local directory
day5-336m-bounded-long	offline	wandb/offline-run-20260913_001732-vk6yw6gm
day5-653m-memory-off	offline	wandb/offline-run-20260913_002221-7jhs2cxf
day5-653m-memory-on	offline	wandb/offline-run-20260913_002238-daxxpskp
day5-336m-preflight	offline	wandb/offline-run-20260913_001703-h62mjhhhy
day5-336m-500m	offline	wandb/offline-run-20260913_010008-c7kbwgbcf/files

6. Limitations

The dataset slice is 10M tokens, the training runs are bounded, the FSDP validation is single-process unless a multi-GPU result is explicitly supplied, and an online W&B; URL is not inferred from offline files. These boundaries should be stated in interviews.